

Canvas

Installation

Prerequisites

- Node.js
- npm
- MongoDB

Clone the repository

```
git clone https://github.com/owen-newberry/ase-285-hw6.git
```

Install Dependencies

```
npm install
```

This will install all of the dependencies in `package.json`, including:

- crypto
- dotenv
- fs
- jest
- mongo
- mongoose

Set Up Environment Variables

Create a `.env` file in the `src` directory and include the following

```
MONGO_URI=your_mongodb_connection_string
```

Using the Application

To use the application, each module will be accessed by the command line.

makepassword.js

To make email and hashed password pairs in password.enc.txt, we simply need to execute this code:

```
cd src  
node makepassword.js
```

This will create a file password.enc.txt, and also upload the encrypted data to MongoDB.

passwordjs.js

To verify email and password pairs, we need to execute this code:

```
node passwordjs.js password.enc.txt [email] [password]
```

This will return either `true` or `false` if the email and password pair match up.

Testing

We have two tests we can execute to ensure our code is working properly.

makepassword.test.js

This will ensure that makepassword.js is creating a file and writing to it, and that our database functionality is working. This uses jest to mock our database, so we aren't in danger of accidentally altering the data.

To run this test, navigate to the root directory:

```
cd ..  
npm test
```

passwordjs.test.js

This test ensures that passwordjs.js is working as intended by executing an acceptance test. To run this, stay in the root directory:

```
npm run acceptance
```

Requirements

- As a user, I should be able to be able to make an encrypted file, password.enc.txt, from passwords.txt.
- As a user, I should be able to upload the encrypted data into MongoDB.
- As a user, I should be able to be given true or false from the software when I give email addresses and passwords.
- As a user, I should be given true from the software if I input a correct email and password pair.
- As a user, I should be given false from the software if I input a correct email and incorrect password.
- As a user, I should be given false from the software if I input a incorrect email and correct password.
- As a user, I should be given false by the software if I input an email not in the password file.
- As a user, I should be given false by the software if I do not input a email.
- As a user, I should be given false by the software if I do not input a password.

Architecture Design

